Examiner reports

Q1.

Most candidates applied the cosine rule correctly, and convincingly obtained the printed result in part (a). Part (b) was beyond the capabilities of many candidates. A significant number tried to use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ without success. Those who quoted the correct identity, $\sin^2\theta + \cos^2\theta = 1$, gained a mark but it was disappointing to see some then take the square root of each term. In general only the better candidates reached the printed answer legitimately. It was pleasing to see many candidates recover to obtain the correct area of triangle ABC.

Q2.

Parts (a) and (b) were very well answered with most candidates scoring full marks. Surprisingly, part (c) proved to be demanding. Many candidates only found 2 rather than 4 solutions and, more alarmingly, candidates seemed to have a problem dealing with the (θ – 0.1) and often subtracted the 0.1, giving an answer of 0.63 rather than 0.83.

Q3.

Candidates usually have difficulty with trigonometry and this session was no exception. Sketching of the graph of $y = \tan x$ was generally not done well and perhaps illustrated the over-reliance on calculators for graphical work. Surprisingly the first branch, between x = 0º and x = 90º, was missing in a significant number of sketches. In part (b) candidates were expected to be able to just write down the two values by first noting by inspection that x = 61 was a solution. Instead many gave the incorrect solutions tan61 and tan241.

Candidates' solutions to show that $\tan\theta = -1$ in part (c)(i) were often unconvincing. The examiners were generally looking for a division by $\cos\theta$ before or after a rearrangement.

Part (c)(ii) starts with the word "Hence' so those candidates who only gave the answers 155º and 335º gained no credit. Those candidates who spotted the link and indicated '$\tan(x - 20°) = -1$' generally went on to gain full marks, although some others subtracted (instead of added) 20 from their answers to part (c)(i). A common error was to start by writing '$\sin(x - 20) + \cos(x - 20) = \sin x - 20 \sin + \cos x - \cos 20$' from which candidates never recovered even after a page of working.

Many candidates gained at least 1 mark for their description of the geometrical transformation in part (d). The final part of the question was answered well by the above average candidates but some left their answer as f($4x$) which was not awarded the mark. A substantial number of candidates gave the answer as 4f(x).

Q4.

(a) Candidates appeared to know the correct trigonometrical identity and this part was well answered.

(b) This was also well answered with candidates able to factorise the expression from part (a).

(c) This was reasonably well answered with most candidates obtaining two correct results and many fully correct answers seen. It was pleasing to see that there were very few candidates who used degrees. They were penalised.

Q5.

This question was answered badly by the majority of candidates. In part (a) the examiners required the word ‘Stretch’ and either the correct direction or the correct scale factor before awarding either of the two marks. It was not uncommon to see the wrong scale factor $\frac{1}{2}$ and the wrong direction (parallel to the y- axis). Many weaker candidates who attempted part (b) started by writing the incorrect statement ‘ x = $\tan^{-1}6$’. Many better solutions were spoiled by manipulation errors illustrated by $\frac{1}{2}x$ = 1.249, 4.3906. $\Rightarrow$ x = 0.6245,2.1958. Part (c) was by far the worst answered part on the paper. Only a very few candidates realised that cos 9 = 0 satisfied the given factorised form of the equation. The vast majority multiplied out the brackets and then cancelled the $\cos\theta$ factor. A mark was awarded to a significant number of candidates for knowledge of the identity ‘ $\tan\theta = \frac{\sin\theta}{\cos\theta}$ ’ in this part of the question.

Q6.

Part (a) Those candidates who appreciated sec x = 1/cos x were usually able to evaluate the first answer of 1.37 radians. When candidates went on to complete this part correctly although 4.51 was a common error from x + 1.37. Answers in degrees were not common but they were seen.

Part (b) This part was answered very well by the majority of candidates when the most appropriate trigonometrical identity was used when candidates changed the equation into one involving sin x and cos x, errors often did occur.

Part (c) Most candidates were able to successfully factorise the quadratic and many went on to complete the question correctly. Marks were lost by ‘extra’ values within the range being given. Candidates were not penalised for extra answers outside the given range.

Q7.

Part (a) Reasonable attempts were made by most candidates.

Part (b)(i) Many candidates answered this correctly but there were many errors such as dividing by 2 first or writing $\sin^{-1} 2x$ as 1/sin $2x$

Part (b) (ii) This part was usually answered correctly. The usual error was the inclusion of a negative sign.

In part (c) although some candidates gave fully correct responses the majority were only able to do the dv 2 first part to get $\frac{dy}{dx} = \frac{2}{\cos y}$.

Q8.

Most candidates were able to state and use the identity 'tan θ = $\frac{\sin\theta}{\cos\theta}$ to obtain the printed result in part (a)(i). The identity '$\cos^2\theta + \sin^2\theta = 1$', required in part (ii), was less well recalled, with sign errors and '$\cos\theta + \sin\theta = 1$' seen. Many candidates correctly solved the equation in part (b)(i) but the explanations given for part (b)(ii) were poor. The most common wrong answer was 'because $\cos\theta = -4$ is negative.' Those who realised the link between the parts normally gained the marks for (b)(iii) and part (c), but many tried to solve part (c) from scratch, despite the clear instruction to 'Write down' the values.

Q9.

Candidates should know the trigonometrical identities as outlined in the specification. Part (a) was well answered and there were many correct solutions to part (b), although some scripts did not give fully convincing arguments. Part (c) required answers correct to the nearest degree, but answers to a greater degree of accuracy were not penalised. There were a significant number of candidates who obtained an answer of 109.47 which was rounded to 109.5 which was then rounded to 110.

Q10.

The great majority of the candidates were able to answer both parts of this question correctly. However, a small number of candidates lost the last accuracy mark which was available for part (b) by not showing enough working. Centres should advise their candidates to show sufficient evidence of working when the answer to a question is given on the question paper.

Q11.

This question was answered very well by the great majority of the candidates. Part (b)(i) of the question was prone to errors for some candidates who did not take sufficient care in getting a correct simplified quadratic equation in tanθ and solving the equation to find the two possible values of θ Some of these candidates wrote 89.8° for a possible value of θ and did not seem to question or check the reasonableness of their answer. For part (b)(ii), almost all candidates chose their smaller angle for the answer, but many could not provide a convincing reason for their choice.

Q12.

This question was answered very well and proved to be an easy source of marks for many candidates.

In part (a)(i), the candidates seemed comfortable with the kinematic equations of motion and the identity $\frac{1}{\cos^2\theta} = 1 + \tan^2\theta$. Very few candidates substituted 20 instead of –20 for y in answering part (a)(ii).

Some candidate lost an accuracy mark in part (b)(i) for premature rounding of the solutions of the equation for tan θ. Other candidates who wrote down inaccurate values for tan θ but used the accurate values to find the values of θ did not lose any accuracy mark. Centres should remind their candidates to give the **final** answer to questions to three significant figures, unless stated otherwise, as stated in the instructions on the front page of the question paper. In part (b)(ii), a small number of candidates gave the result to less than three significant figures.

Q13.

Parts (a), (b) and (c)(i) of this question were answered well by many candidates. However, some candidates lost one or more accuracy marks for parts (b) and (c)(i). To answer part (c)(i), most candidates evaluated 10 cos 26.9° and 10 cos 74.4° to arrive at the correct conclusions. A small number of candidates stated that the can will be knocked off the wall if 10 cos α > 8, or cos α > 0.8, and then thought that this would imply α > 36.9°.

Q14.

The responses to part (a) were very good. The vast majority of candidates were familiar with the kinematics equations for the motion of a projectile.

Many candidates answered part (b)(i) of the question correctly. However, some candidates misunderstood the axes and substituted –1 instead of 1 for y. Evidently, because the answer was given, some of these candidates were able to rectify their mistakes but others gave wrong answers.

The candidates understood and used the condition for real solutions for the quadratic equation to arrive at the relationship requested for part (b)(ii). For part (b)(iii), almost all candidates substituted 5 for R. However, some candidates were unable to proceed further and solve the quadratic equation in u^2 in order to find the minimum speed of projection. There were other candidates who attempted to use differentiation to find the minimum speed. These candidates tried to minimize the expression $u^4 - 20u^2 - 2500$ rather than finding the minimum value of u.

Q15.

Part (a) The vast majority of candidates were familiar with the equations of motion for a projectile. They were able to eliminate t and arrive at the required equation of the trajectory.

Part (b) The algebraic manipulation and 'tidying up' of the quadratic equation in x and the subsequent task of solving it were prone to errors for some candidates. Some candidates who solved the equation correctly did not identify the required answer from their two solutions.

Part (c) Most candidates stated the two assumptions; no air resistance and the ball is a particle. A small minority of the candidates mistakenly stated that the ball was a particle without that mass.

Q16.

The vast majority of the candidates answered part (a) of the question correctly.

For part (b), almost all candidates substituted 100 for x and 4 for y. However, some candidates had difficulty simplifying the resulting equation.

For part (c), most candidates stated that they assumed the ball was a particle, there was no air resistance or that the only force acting on the ball was the Earth's gravitational force. However, there were some candidates who stated the assumption of the ball being a particle without mass!